PLANET DEFENSE

Global Climate Intelligence

Regional Climate Risk Scoring Methodology

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CONFIDENTIAL

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Executive Summary

Planet Defense's Regional Climate Risk Scoring System provides a comprehensive, forward-looking assessment of climate-related risks at the sub-regional level. This methodology evaluates 27 global sub-regions across six continents, incorporating both physical climate hazards and economic transition risks.

The scoring system synthesizes five critical dimensions of climate risk into a single, actionable Overall Risk Score ranging from 0 to 100, where higher scores indicate greater risk exposure. A key innovation is the integration of GDP per capita as a vulnerability amplifier, recognizing that lower-income regions face disproportionate impacts from the same climate hazards due to limited adaptation capacity and financial resilience.

This holistic approach enables investors, policymakers, and business leaders to identify risk concentrations, compare exposures, assess preparedness gaps, and monitor progress. The methodology is grounded in internationally recognized frameworks including TCFD, ISSB, and SBTi, while adding regional granularity that industry-level analysis cannot capture.

1. Introduction

1.1 Purpose and Scope

Climate change presents both physical hazards and economic transition risks that vary dramatically by geography. This methodology provides a systematic framework for quantifying regional climate risk across 27 sub-regions spanning six continents. Each assessment integrates physical climate science, economic transition modeling, policy analysis, and financial capacity evaluation.

1.2 Geographic Coverage

The methodology covers 27 sub-regions organized by continent:

- North America: United States, Canada, Mexico & Central America, Caribbean
- South America: Brazil, Southern Cone, Andean Region, Guianas
- Europe: Western Europe, Northern Europe, Southern Europe, Eastern Europe
- Africa: North Africa, West Africa, East Africa, Central Africa, Southern Africa
- Asia: East Asia, Southeast Asia, South Asia, Central Asia, Middle East
- Oceania: Australia, New Zealand, Melanesia, Polynesia, Micronesia

1.3 Temporal Horizons

Risk projections extend to 2030 (near-term strategic planning) and 2050 (long-term aligned with net-zero targets and Paris Agreement goals).

2. Methodology Overview

2.1 The Five-Pillar Approach

Regional climate risk is assessed across five interconnected dimensions scored 0-100. The Overall Risk Score combines all five using calibrated weights, with GDP per capita serving as a vulnerability amplifier for physical and transition risk components.

Physical Risk Score (25% weight)

Quantifies exposure to direct physical climate impacts including temperature increase, extreme weather, water stress, and sea level rise. GDP per capita vulnerability multiplier applied.

Transition Risk Score (35% weight)

Measures economic exposure to the low-carbon transition including carbon pricing, stranded assets, regulatory gaps, and fossil fuel dependency. GDP per capita vulnerability multiplier applied.

Transition Readiness Score (20% weight, inverted)[1]

Evaluates preparedness through emission reduction progress, renewable energy adoption, climate investment, and technology deployment. Higher readiness reduces overall risk.

Financial Resilience Score (15% weight, inverted)

Assesses financial capacity through green economy growth, carbon cost burden, and access to green finance. Higher resilience reduces overall risk.

Adaptation Score (5% weight, inverted)

Measures adaptation progress through infrastructure resilience investment and resource efficiency improvements. Higher adaptation reduces overall risk.

2.2 GDP per Capita Vulnerability Multiplier

GDP per capita acts as a vulnerability amplifier, recognizing that lower-income regions face disproportionate climate impacts due to limited adaptation capacity, weaker infrastructure, reduced financial buffers, and constrained institutional capabilities.

The multiplier formula:

$$\text{Vulnerability Multiplier} = 1 + [(50{,}000 - \text{GDP per capita}) / 100{,}000]$$

The multiplier adjustment is applied only to the first and second components in 2.1.

2.3 Overall Risk Score

The overall risk score is the weighted sum of the five components, post-adjustment, as described in 2.1-2.2 above.

3. Risk Rating Bands

The Overall Risk Score (0-100) is categorized into six rating bands:

[1] Throughout this report, when calculating scores, inverted components use the 0-100 complement of its score.

Score Range	Rating	Description
0-20	**Minimal Risk**	Very low climate risk with strong preparedness
21-35	**Low Risk**	Manageable risks with good preparedness
36-50	**Moderate Risk**	Balanced profile requiring monitoring
51-65	**Elevated Risk**	Significant risks requiring action
66-80	**High Risk**	Substantial risks requiring urgent action
81-100	**Critical Risk**	Extreme risks with major deficiencies

4. Regional Risk Profiles and Case Studies

The following profiles illustrate how the methodology operates across diverse geographic and economic contexts, highlighting the interplay between physical exposure, transition pressures, and adaptive capacity.

4.1 Northern Europe: Low Risk Through Proactive Preparation

Risk Profile: Low to Minimal Risk (typically 25-35 overall score)

Northern Europe demonstrates how early climate action and robust institutions can offset moderate physical risks. Despite facing increasing temperatures and changing precipitation patterns, the region benefits from high GDP per capita that reduces vulnerability. Strong renewable energy deployment (often exceeding 50% by 2030 projections), ambitious emission reduction commitments, and substantial climate adaptation investments drive high readiness and resilience scores. The region's diversified economy and access to green finance further buffer transition risks. Key success factors include long-standing carbon pricing mechanisms, public investment in clean infrastructure, and social consensus on climate action. This profile illustrates that wealth combined with political will creates the lowest-risk climate trajectory.

4.2 Caribbean: High Physical Risk Despite Mitigation Efforts

Risk Profile: Elevated to High Risk (typically 55-70 overall score)

The Caribbean exemplifies regions where extreme physical exposure dominates the risk profile despite meaningful adaptation efforts. Small island states face existential threats from sea level rise, hurricane intensification, and coastal erosion. The GDP per capita vulnerability multiplier amplifies these physical risks significantly. While many Caribbean nations show commitment to renewable energy transition and adaptation planning, limited financial resources constrain implementation scale. Tourism-dependent economies face both physical disruption risks and transition challenges as aviation decarbonizes. This profile demonstrates that geographic vulnerability can create high risk scores even with strong policy intentions, highlighting the need for international climate finance and loss-and-damage mechanisms.

4.3 Middle East: Transition Risk from Economic Structure

Risk Profile: Moderate to Elevated Risk (typically 50-65 overall score)

The Middle East presents a transition risk-dominant profile driven by fossil fuel economic dependence. Despite relatively high GDP per capita in many sub-regions, the vulnerability multiplier still applies due to concentration risk in hydrocarbon sectors. Stranded asset exposure is severe, with substantial proven reserves that may become unburnable under Paris-aligned pathways. Extreme physical risks compound challenges, particularly water stress and heat exposure. However, some Gulf states demonstrate rapid renewable energy deployment and economic diversification efforts, creating variance within the region. This profile illustrates how economic structure can create transition vulnerability even in wealthy regions, and how diversification is essential for climate resilience.

4.4 Sub-Saharan Africa: Compound Vulnerabilities

Risk Profile: High to Critical Risk (typically 65-85 overall score)

Sub-Saharan African regions often face the highest risk scores due to compound vulnerabilities amplified by low GDP per capita. Severe physical exposure (droughts, flooding, agricultural disruption) combines with limited financial capacity for adaptation. The GDP vulnerability multiplier substantially increases both physical and transition risk scores. While the region has minimal historical emissions responsibility, it faces transition challenges

through stranded fossil fuel assets in some areas and need for rapid clean energy access expansion. Low readiness and resilience scores reflect infrastructure and institutional constraints rather than lack of ambition. This profile underscores climate injustice: regions least responsible for emissions face disproportionate impacts with least capacity to respond. It highlights the critical importance of adaptation finance and technology transfer.

5. Data Sources and Quality Assessment

5.1 Climate Science Data

Temperature and Physical Risk Projections:

- Source: CMIP6 climate model ensemble (Coupled Model Intercomparison Project Phase 6)
- Scenario: SSP2-4.5 (middle-of-the-road pathway) as baseline
- Resolution: Regional downscaling for sub-continental accuracy
- Update Frequency: Annually as IPCC assessment cycles progress
- Quality: High confidence for temperature, medium for precipitation patterns

Water Stress Analysis:

- Source: World Resources Institute Aqueduct platform
- Methodology: Basin-level water supply/demand modeling
- Update Frequency: Every 2-3 years

Sea Level Rise and Coastal Exposure:

- Source: IPCC AR6 sea level projections, NASA sea level portal
- Coastal asset mapping: OpenStreetMap, regional infrastructure databases
- Quality: High confidence for global mean, medium for regional variations

5.2 Economic and Emission Data

GDP and Economic Indicators:

- Sources: World Bank World Development Indicators, IMF World Economic Outlook
- GDP per capita: Purchasing power parity adjusted, 2024 baseline
- Update Frequency: Annual
- Quality: High for developed regions, medium for emerging markets

Emission Trajectories and Projections:

- Historical: EDGAR (Emissions Database for Global Atmospheric Research)
- Projections: IEA World Energy Outlook, NGFS climate scenarios
- Scenario: Current policies baseline with stated policy variants
- Update Frequency: Annual

Carbon Pricing and Policy:

- Sources: World Bank Carbon Pricing Dashboard, ICAP ETS database
- Policy tracker: Climate Action Tracker, NewClimate Institute
- Update Frequency: Quarterly for prices, annually for policy assessments

5.3 Energy and Technology Data

Renewable Energy and Infrastructure:

- Sources: IRENA renewable capacity statistics, IEA energy statistics
- Technology deployment: Global CCS Institute, regional industry associations
- Update Frequency: Annual
- Quality: High for OECD countries, medium for emerging markets

5.4 Data Quality by Region

Data confidence varies systematically by region:

High Confidence Regions: North America, Western Europe, Northern Europe, Australia, New Zealand, East Asia (Japan, South Korea). Comprehensive monitoring, frequent updates, validated models.

Medium Confidence Regions: Southern Europe, Eastern Europe, Southern Cone, Middle East, Southeast Asia. Good coverage with some gaps in subnational data or less frequent updates.

Lower Confidence Regions: Sub-Saharan Africa, Central Asia, Pacific Islands. Data gaps in historical records, less climate model validation, institutional capacity constraints. Scores in these regions carry larger uncertainty bounds.

5.5 Validation and Calibration

The methodology undergoes annual validation through comparison with observed climate impacts, realized policy changes, and stakeholder feedback. Scoring algorithms are recalibrated when systematic biases emerge. Regional experts review scores for their areas to ensure alignment with on-the-ground conditions. This iterative process ensures the methodology evolves with improving data quality and scientific understanding.

Appendix A: GDP per Capita by Sub-Region

The following table lists the GDP per capita values (PPP-adjusted, 2024 USD) used in calculating the vulnerability multiplier for each sub-region:

Sub-Region	GDP per Capita (USD)
United States	76,421
Canada	54,823
Mexico & Central America	12,000
Caribbean	18,125
Brazil	9,803
Southern Cone	15,004
Andean Region	8,533
Guianas	7,040
Western Europe	52,452
Northern Europe	65,017
Southern Europe	35,003
Eastern Europe	19,220
North Africa	6,532
West Africa	2,890
East Africa	3,233
Central Africa	2,086
Southern Africa	7,543
East Asia	28,046
Southeast Asia	11,058
South Asia	2,405
Central Asia	9,021
Middle East	23,014
Australia	62,000
New Zealand	48,000
Melanesia	4,221

Sub-Region	GDP per Capita (USD)
Polynesia	8,517
Micronesia	5,317

Note: *GDP per capita values are PPP-adjusted 2024 USD. Regional aggregates represent population-weighted averages where applicable. Data sourced from World Bank World Development Indicators and IMF World Economic Outlook.*

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